Defensible Survival Extrapolation in R

A practical map of statistical approaches and R tools

Organized as a sequence of analytical decisions, grounded in NICE TSD14[1].





Systematic Workflow Overview

NICE TSD14 recommends a structured four-step procedure to ensure that survival extrapolations are both statistically sound and clinically credible. Each step builds on the last, guiding the analyst from exploratory data inspection through to a fully justified model selection that can withstand scrutiny in a health technology appraisal.

Step 1 — Assess Model Assumptions & Hazard Structure

Examine the observed data using graphical and statistical tools to understand what survival and hazard shapes are plausible **before** selecting any model.

Step 2 — Fit Candidate Models

Fit the full standard parametric family (Exponential, Weibull, Gompertz, Log-normal, Log-logistic, Generalized Gamma) and flexible alternatives where justified.

Step 3 — Evaluate Statistical Fit & Graphical Diagnostics

Compare fitted curves against the Kaplan-Meier estimate using AIC/BIC and residual diagnostics. Visually inspect hazard plots, probability plots, and residual plots to detect systematic departures.

Step 4 — Evaluate External Validity & Decision Uncertainty

Assess whether the extrapolated curves are clinically plausible, consistent with external evidence, and robust to alternative model choices.



Step 1: Assess Model Assumptions & Hazard Structure

Before selecting a distribution, understand what the observed data can support. The question is not *"Which model has the lowest AIC?"* – it is **"What survival and hazard structures are reasonably supported by the data?"**

Log-Cumulative Hazard Plots

`survival` → `survfit()` – Explore hazard behaviour and assess proportional hazards.

Q-Q-Type Plots for Censored Data

`GofCens` → `probPlot()` – Examine compatibility with distributional assumptions.

Residual Diagnostics

`flexsurv / survival` – Cox-Snell residuals identify systematic lack of fit.



Steps 2 & 3: Fit Candidates & Evaluate Internal Fit

Fit Candidate Models

TSD14 recommends considering the full standard parametric family via `flexsurv::flexsurvreg()`:

Exponential → Weibull → Gompertz → Log-normal → Log-logistic → Generalized Gamma

When additional flexibility is justified: `rstpm2::stpm2()` or piecewise approaches via `survival::survSplit()`. Greater flexibility does **not** automatically mean better extrapolation.

KM vs Fitted Curves

Does the model reproduce observed survival? `flexsurv::plot()`

AIC / BIC

Do some distributions provide substantially better statistical fit?

Residual Diagnostics

Systematic departures? A lowest-AIC model can still yield an implausible long-term curve.





Step 4: Assess Extrapolation Plausibility

This is where survival modelling becomes **decision analysis**, not simply curve fitting.

Clinical Plausibility

Is the long-term hazard clinically reasonable?

External Evidence

Does predicted survival align with external data and registries?

Biological Reasoning

Are differences between treatment arms biologically reasonable?

Sensitivity Analysis

How sensitive are lifetime outcomes to alternative plausible models?

Summary: Defensible Survival Extrapolation Workflow

Step	Statistical Method	R Tool	Decision
Step 1: Assess Model Assumptions & Hazard Structure	Log-Cumulative Hazard Plots	`survival` → `survfit()`	Are proportional hazards reasonable? What hazard shape is supported?
Step 1: Assess Model Assumptions & Hazard Structure	Q-Q-Type Plots for Censored Data	`GofCens` → `probPlot()`	Which distributional families are compatible with the data?
Step 1: Assess Model Assumptions & Hazard Structure	Residual Diagnostics	`flexsurv` / `survival` (Cox-Snell residuals)	Is there systematic lack of fit in any candidate?
Step 2: Fit Candidate Models	Standard Parametric Family	`flexsurv::flexsurvreg()`	Fit Exponential, Weibull, Gompertz, Log-normal, Log-logistic, Generalized Gamma
Step 2: Fit Candidate Models	Flexible / Piecewise Models	`rstpm2::stpm2()` / `survival::survSplit()`	Is additional flexibility statistically and clinically justified?
Step 3: Evaluate Internal Fit	KM vs Fitted Curves	`flexsurv::plot()`	Does the model reproduce observed survival within the trial period?
Step 3: Evaluate Internal Fit	Information Criteria	AIC / BIC (from `flexsurvreg` output)	Do some distributions provide substantially better statistical fit?
Step 3: Evaluate Internal Fit	Residual Diagnostics	`flexsurv` / `survival`	Are there systematic departures? Lowest AIC ≠ best extrapolation.
Step 4: Assess Extrapolation Plausibility	Clinical Plausibility Review	Expert judgment + external data	Is the long-term hazard clinically reasonable?
Step 4: Assess Extrapolation Plausibility	External Validation	Registry / published survival data	Does predicted survival align with external evidence?
Step 4: Assess Extrapolation Plausibility	Sensitivity Analysis	Multiple model comparison	How sensitive are lifetime outcomes to alternative plausible models?

